Diagnostic

Name ________________________________ **Course** BioE 147 / 247 *(circle one)*

This isn't graded and doesn't touch your grade. It's here so I can pitch the next few weeks at *you* rather than at an imaginary average student.

The form told me what you've used. These four are the other half — what you can do right now, cold. They come from four different backgrounds on purpose, and **almost nobody answers all four.** If you haven't seen something, leave it blank: a blank tells me something a guess doesn't, and it costs you nothing. Please don't look anything up — if you do, I'll cheerfully skip past something you'd rather I explained.

1. Molecular

A gene is preceded by a promoter. A protein called a *repressor* binds a site that overlaps that promoter.

In one or two sentences: what physically stops the gene being expressed, and what would happen to expression if you doubled the number of repressor molecules in the cell?

2. Dynamical

A protein is made at a constant rate α (molecules per minute) and each molecule is removed with probability per unit time γ, so

$$\frac{dx}{dt} = \alpha - \gamma x.$$

(a) What is the steady-state value of x?

(b) Starting from $x = 0$, roughly how long does it take to get most of the way there? Give the answer in terms of α and γ.

3. Physical

An *E. coli* cell is about **1 femtolitre** (10^{-15} L).

Approximately how many molecules are in that cell at a concentration of **1 nM**? (Avogadro's number is 6×10^{23} mol^{-1}.) An order of magnitude is a complete answer; show whatever working you use.

4. Computational

```
import numpy as np

def f(v):
    return np.array([v[1], -v[0]])

x = np.array([1.0, 0.0])
for _ in range(3):
    x = x + 0.1 * f(x)
print(x)
```

What does this print, approximately? And what is this code doing?

Leave this with me on your way out. Nothing here goes near the gradebook — I read them, I work out what to do differently, and that's the end of it. If a question looked like a foreign language, that's fine, and it's exactly what I want to know.